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Unit – 1 : Session – 6 : SLO – 2 | Complement arithmetic

■ MCQs

1. c — 0110 ($1010 \rightarrow 0101 + 1$)
2. c — 2's complement
3. b — 0011
4. c — Take 1's complement and mark result negative
5. d — Sign of the number

■ Two-Mark Questions

1. 1's complement of a binary number is obtained by inverting every bit ($0 \rightarrow 1$, $1 \rightarrow 0$). For a 4-bit number, e.g. $1011 \rightarrow 0100$. It represents the negative of the number in 1's complement arithmetic.
2. 2's complement subtraction rules: (i) find 2's complement of the subtrahend; (ii) add it to the minuend; (iii) if a carry is generated, discard it — result is positive and final; (iv) if no carry, result is negative — take 2's complement again for the magnitude and attach '-'.

■ Scenario-Based Long Answer

- $A - B = 1011 - 0101$ using 1's complement: 1's complement of $B = 1010$.
- Add: $1011 + 1010 = 1\ 0101$. End-around carry present $\rightarrow$ add it back to the LSB: $0101 + 1 = 0110$.
- Final result: $0110_{10} = 6_{10}$ (positive, correct: $11 - 5 = 6$). Had there been no carry, the result would be negative and interpreted by taking its 1's complement.

■ Match the Following

1. 1's Complement — a) Invert all bits ($0 \rightarrow 1$, $1 \rightarrow 0$)
2. 2's Complement — b) Add 1 to 1's complement
3. Carry in 2's complement — c) Ignore and result is positive
4. No carry in 1's complement — d) Take 1's complement and add '-'
5. Sign-magnitude representation — e) MSB for sign, remaining bits for value

■ Fill in the Blanks

1. 1000 ($0111+1$)
2. negative (2's complement form)
3. 1111
4. Most Significant Bit
5. 1's complement

■ Scenario-Based Small Activity

- $A - B = 0101 - 0011$. 2's complement of $B = 1100 + 1 = 1101$.
- Add: $0101 + 1101 = 1\ 0010 \rightarrow$ discard carry $\rightarrow 0010$.
- Final answer: $0010_{10} = 2_{10}$ ($5 - 3 = 2$).

■ True or False

1. FALSE — Flipping bits gives 1's complement; add 1 for 2's.
2. FALSE — 2's complement is used today.

3. TRUE 4. TRUE 5. TRUE